

CO-1 Output Prediction

Q1:

 **SIMATS**
ENGINEERING

SIMATS Online Programming Lab
DS PROGRAMMING

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Q1
Searching
1 mark

Q2
Array
1 mark

Q3
Merging
1 mark

3/3 answered

Write a C program for Merge Two Arrays

Your Code

```
1 #include<stdio.h>
2 void main()
3 {
4     int arr1[]={1,2,3};
5     int arr2[]={4,5,6};
6     int n1=sizeof(arr1)/sizeof(arr1[0]);
7     int n2=sizeof(arr2)/sizeof(arr2[0]);
8     int merged[n1+n2];
9     for (int i=0;i<n1;i++){
10         merged[i]=arr1[i];
11     }
12     for(int i=0;i<n2;i++){
13         merged[n1+i]=arr2[i];
14     }
15     printf("merged array:");
16     for (int i=0;i<(n1+n2);i++){
17         printf("%d",merged[i]);
18     }
19 }
```

Submitted

Input
stdin input

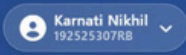
Output
merged array:123456

Q2:



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Q1 Searching 1 mark

Q2 Array 1 mark

Q3 Merging 1 mark

3/3 answered

Write a program for binary search

Your Code

```

1 #include<stdio.h>
2 int binarySearch(int arr[],int size,int target)
3 {
4     int low=0;
5     int high=size-1;
6     while(low <= high)
7     {
8         int mid = low + (high-low)/2;
9         if (arr[mid]==target){
10             return mid;
11         }
12         if (arr[mid] < target){
13             low = mid +1;
14         }
15         else {
16             high=mid-1;
17         }
18     }
19     return -1;
20 }
21 int main(){
22     int arr[]={12,23,34,45,67,78};
23     int target = 45;
                
```

Input

45

Output

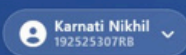
element 45 found at index 3.

Q3:



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Q1 Searching 1 mark

Q2 Array 1 mark

Q3 Merging 1 mark

3/3 answered

Write a C program for Count the Occurrences of an Element

Your Code

```

1 #include<stdio.h>
2 void main()
3 {
4     int arr[10],n,i,count=0,element;
5     printf("enter the size of an array :");
6     scanf("%d",&n);
7     printf("enter the no of elements in the array :");
8     for(i=0;i<n;i++){
9         scanf("%d",&arr[i]);
10    }
11    printf("enter the element to find :");
12    scanf("%d",&element);
13    for(i=0;i<n;i++)
14    {
15        if(arr[i]==element)
16        {
17            count++;
18        }
19    }
20    printf("%d occurs %d times.\n",element,count);
21
22 }
                
```

Input

4
2 3 3 3
3

Output

enter the size of an array :enter
the no of elements in the array
:enter the element to find :3
occurs 3 times./n

Submitted